(13) Check output bearing preload. **Output bearing preload must be checked and/or adjusted if any of the following items have been replaced:**

- Output shaft (rear carrier assembly)
- Output shaft bearings
- Transmission case

(a) **PRELOAD CHECK/SHIM SELECTION:** Install rear output shaft bearing cone and special tool 6618A (Fig. 92).

(b) Install special tool 6618A (Fig. 93). Lightly tighten retaining screws. Screws should be below the plate surface, but do not snug screws.

(c) Turn case over on arbor press so that the plate is resting on the press base. **CAUTION: The output shaft will extend through the hole of tool 6618A. Ensure your press table has clearance for the output shaft.**

(d) Install shim on output shaft (Fig. 94). Apply small amount of petrolatum onto the shim to hold it in place. Use the original shim as a starting point. If original shim is not available, use the thickest shim available.

(e) Install output shaft/rear carrier into rear bearing. The shaft must be pressed into position. Use special tool MD-998911 (Disc) and C- 4171 and C4171-2 (Handle) to press shaft into rear bearing (Fig. 95).

(f) **Do not re-use old output shaft nut because the removed stake weakens the nut flange.** Using special tools 6497 and 6498-A, install new output shaft nut. Do not reuse old output shaft nut. Tighten new output shaft nut to 271 N·m (200 ft. lbs.).

(g) Check the turning torque of the output shaft (Fig. 96). The shaft should have 1 to 8 in. lbs. of turning torque. If the turning torque is **higher than** 8 in. lbs., install a thicker shim. If turning torque is **less than** 1 in. lb., install a thinner shim. Make sure there is no end play.

(h) The new nut must be staked after the correct turning torque is obtained (Fig. 97) (Fig. 98). Use special tool 6639 to stake output shaft nut. **CAUTION: Failure to stake nut could allow the nut to back-off during use.**

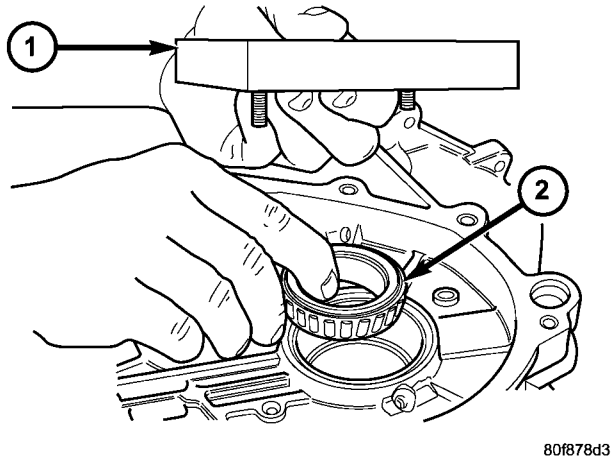

Fig. 92 Bearing Installation

1 - SPECIAL TOOL 6618-A
2 - REAR OUTPUT SHAFT BEARING

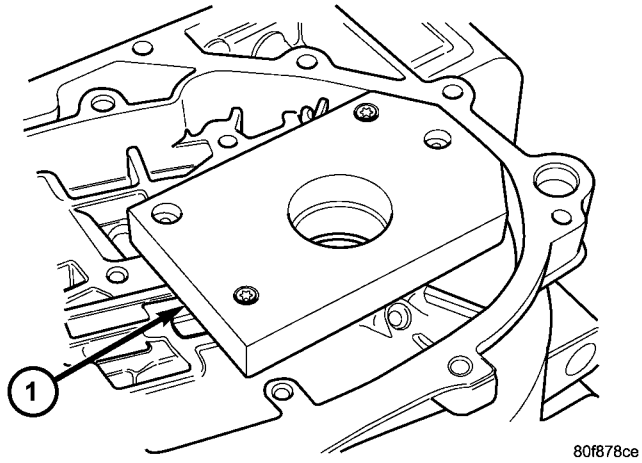

Fig. 93 Special Tool Installed

1 - SPECIAL TOOL 6618-A